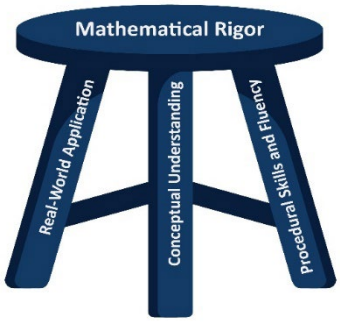
Shifts in the 2022 Academic Standards in Mathematics

The 2022 Minnesota Academic Standards in Mathematics will be fully implemented in the 2027-28 school year. Information and resources can be found on the Minnesota Department of Education Math Standards Implementation page.



The Career, College and Community Readiness Vision Statement describes the mathematical experiences all Minnesota K-12 students should have to prepare them for life after graduation.

The vision for Minnesota's math classrooms can be represented by a three-legged stool, with each leg symbolizing an essential component of mathematical rigor: conceptual understanding, real-world application, and procedural skills and fluency.

This document describes shifts from the 2007 to 2022 Minnesota Mathematics Standards to support districts and educators in instructional decision-making. The shifts are addressed at three levels: overall highlights, shifts within each of the seven K-12 anchor standard progressions (both structural and content related), and content changes (both omissions and additions) by grade level.

Overall Highlights

Structural Shifts

- The number of strands was reduced from four to three in the 2022 Standards. The strands represent the reporting categories for the MCAs.
- The seven anchor standards represent K-HS math progressions with a collection of benchmarks at each grade.
- The symbols at the end of each benchmark represent the 2 dimensions, math practices and benchmark contexts, that overlay the teaching and learning of mathematics.

Content Shifts

- The Career, College and Community Readiness Vision Statement defines what the mathematics student will need for effective participation in society, including in their career(s), post-secondary education, and daily decision-making.
- The standards emphasize mathematical rigor through equal focus on procedures & skills, concepts, and application.
- There is a greater proportion of data and probability in the 2022 Standards.

Shifts in K-12 Anchor Standard Progressions

Strand 1: Data and Probability

"*Data are not merely numbers but numbers in context*" (Cobb and Moore, 1997). Students will develop questions about situations impacting their everyday lives, make a plan to produce data and provide answers by organizing, describing and summarizing the data. In addition, students will develop and evaluate inferences and make predictions. They will find patterns and order from random phenomena to analyze and understand important relationships and use the concept of variability to make sense of the world.

Anchor Standard 1: Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data, and communicating the results.

The 2020 Guidelines for Assessment and Instruction in Statistics Education II (GAISE II) states, "The digital revolution has made data readily accessible to statistical methods and technological tools so that students can gain insights and make recommendations to manage pressing world issues." Analyzing and interpreting data requires critical thinking, creativity, and a nuanced understanding of the data's context. These are the mathematical skills Minnesota students need to be career, college, and community ready.

Structural Shifts

- Data is the first strand in the 2022 Math Standards. The committee made an intentional decision to move data from last position in the 2007 Standards to emphasize its importance and signify it as a priority.
- Data Science benchmarks were written with the guidance of the national GAISE statistical standards.

Content Shifts

- The Data Science standard includes four K-12 learning progressions, each aligned to a stage of the data cycle: (1) Formulate Statistical Investigative Questions, (2) Collect and Consider the Data, (3) Analyze the Data, and (4) Interpret Results. These progressions are reflected in the benchmark level language.
- Data Science benchmarks are new in grades K-2.
- The 2022 Standards significantly expand Data and Probability at the high school level - approximately one-third of the high school math standards are in this strand. High schools will want to review their 9-11th grade course sequences to determine where these benchmarks best fit.
- Data Science benchmarks are designed to be taught contextually (Dimension 2) as part of the four-part Data Cycle.

Strand 1: Data and Probability

"*Data are not merely numbers but numbers in context*" (Cobb and Moore, 1997). Students will develop questions about situations impacting their everyday lives, make a plan to produce data and provide answers by organizing, describing and summarizing the data. In addition, students will develop and evaluate inferences and make predictions. They will find patterns and order from random phenomena to analyze and understand important relationships and use the concept of variability to make sense of the world.

Anchor Standard 2: Chance and Uncertainty

Apply and explain the concepts of probability to interpret data, generate questions, predict and make informed decisions to solve problems and communicate ideas.

A greater focus on data literacy in the standards necessitates the development of probabilistic thinking. "To adequately function in society, citizens need to overcome their deterministic thinking and accept the existence of fundamental chance in nature." Anchor Standard 2 develops students critical thinking about the meaning of chance and applies the concept of uncertainty in real life.

Structural Shifts

- Probability (Chance & Uncertainty) was separated from data science.
- Chance and Uncertainty (probability) is a new topic in grades 1-5.
- Anchor Standard 2 has no benchmarks in kindergarten or grade 8.

Content Shifts

- Students will find patterns and order from random phenomena (note: this term aligns with the 2018 MN Science Standards).
- Students use the concept of variability to make sense of the world.
- Students will develop probability vocabulary in the early grades so they can begin to think critically about the meaning of chance. Concepts can be introduced through games, simulations, and other contexts involving random phenomena.
- Students will formalize their understanding in the later grades and apply probability concepts to make predictions and justify their reasoning.
- Students will be given multiple chances to quantify and describe uncertainty.
- Students will focus on applying probability concepts to decision making, financial readiness, and problem solving in real-world situations such as health decisions, weather forecasts, investing money, and analyzing risks. (Dimension 2)

Strand 2: Spatial Reasoning

Students will actively make sense of relationships between and within geometric figures, generalize statements about the figures and develop arguments about these relationships.

Anchor Standard 3: Measurement

Investigate measurement using a variety of tools, units, systems, processes, and techniques in various cultures. Explain and reason with attributes, estimations, and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.

Geometry means 'earth measure,' connecting measurement to spatial reasoning. Students use measurement daily to explore questions about their school or home environments. The measurement standard represents the big idea that objects have measurable attributes that can be quantified using units.

"Measurement is a foundational idea in mathematics and science, yet it is often overlooked in importance. We must help students make sense of it, not just compute it." – Douglas Clements

Structural Shifts

- Measurement is separate from Geometry in the 2022 Standards. The measurement standard focuses on quantifying attributes of objects using units, while the Geometry standard focuses on analyzing shapes, their properties, and their relationships to make mathematical arguments and solve problems.
- Math Practice 6: Attend to Precision is critical, from specifying units to rounding, depending on context.
- Second Grade has the largest increase in measurement benchmarks.

Content Shifts

- Students identify attributes that can be measured.
- Students identify progressions of measurement in money, and in one-dimensional, two-dimensional, and three-dimensional measurements.
- Students investigate measurement using various tools, units, systems, and cultural practices, with emphasis on estimation, justification, and determining reasonableness.
- Students will use measurable or quantifiable attributes: length, height, weight, perimeter, area, surface area, volume, capacity, angles, time, temperature, density, and money.
- The 2022 Standards to not include telling time in 1st-3rd grade benchmarks. However, schools can use time as a context in learning math concepts such as skip counting by fives, equivalence (numeral 3 represents 15 minutes past the hour), addition and subtraction with elapsed time, modular arithmetic, fractions, and circles (degrees and angles).

Strand 2: Spatial Reasoning

Students will actively make sense of relationships between and within geometric figures, generalize statements about the figures and develop arguments about these relationships.

Anchor Standard 4: Geometry

Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems, and communicate ideas.

Geometry is the study of shapes, positions, and the relationships between objects in space.. Students use geometric figures and visual reasoning to explore and understand the world around them. The word "geometry" comes from the Ancient Greek words *geo* (earth) and *metron* (measurement). Geometry connects mathematical concepts to students' lived experiences by helping them measure and understand the physical world.

Structural Shifts

- The strand language calls for students to actively make sense of relationships, generalize statements, and develop arguments.
- Students are asked to communicate geometric ideas.
- Some definitions have changed. For example, trapezoid now uses an inclusive definition: a quadrilateral with at least one pair of parallel sides (rather than exactly one pair). MCA-IV items will accept either definition. Refer to the Minnesota Math Educator Glossary for more information.

Content Shifts

- Students will move through stages of geometric learning: beginning with visualization and identifying classes of shapes, progressing to analyzing properties and understanding relationships, and ultimately engaging in formal deductive reasoning.
- Students use visuals and representations (such as drawings) to explain and justify their thinking.
- In fourth grade, the 2022 Standards focus on parallel and perpendicular lines (moved from third grade). Translation, reflection, and rotation are omitted from fourth grade.
- The 2022 Standards place greater emphasis on orientation and relative position of figures.
- In high school, students apply concepts of density based on area and volume in contextual situations.

Strand 3: Patterns and Relationships

Mathematics is the science of pattern and order. Students will organize and describe the world using representations of numbers and operations. They will learn actions that transform numbers as well as ways of thinking that bring them back to where they started. Students will analyze and describe relationships among variables by focusing on how variables change and stay the same.

Anchor Standard 5: Number Relationships

Describe, interpret and use quantities, relationships between quantities, representations of quantities, and number systems. The student will describe operations and the relationship between operations. The student will use strategies and procedures accurately, efficiently, and flexibly. Assess the reasonableness of the results.

The standard connects knowledge of numbers to operations estimation and applies this to real-world situations. In addition, students demonstrate an understanding of numerical relationships, properties of numbers and operations, generalize from numerical patterns, and verify results.

Structural Shifts

- Anchor Standard 5 has the largest number of benchmarks with 117 of 373 benchmarks or approximately one-third of all benchmarks in the standards.
- There is an increased emphasis on explaining and justifying throughout the grade levels.
- The inclusion of Dimension 1 (Mathematical Practices) and Dimension 2 (Benchmark Contexts) encourage integration of numeracy benchmarks with mathematical processes and contextual applications.

Content Shifts

- The standard is separated into four progressions: (1) quantities, (2) describing operations and relationships, (3) use strategies and procedures, and (4) assess reasonableness.
- Anchor standard 5 identifies many strategies and representations to be used in the conceptual development of number relationships.
- In K-5 there is an increased focus on using groups of ten and place value language. Fraction development begins in first grade and is more detailed in benchmarks.
- In grades 6-High School, students apply number relationships to the five benchmark contexts with special emphasis on financial literacy.
- Grade 6 now includes negative rational numbers (moved from grade 7).
- High school now includes scientific notation, rational exponents, composition of functions, and matrices.
- The definition of fluency includes accuracy, efficiency, and flexibility.

Strand 3: Patterns and Relationships

Mathematics is the science of pattern and order. Students will organize and describe the world using representations of numbers and operations. They will learn actions that transform numbers as well as ways of thinking that bring them back to where they started. Students will analyze and describe relationships among variables by focusing on how variables change and stay the same.

Anchor Standard 6: Equivalence and Relational Thinking

Use concepts and properties of equivalence and relational thinking to represent and compare numerical expressions, proportional relationships, algebraic expressions, and equations.

"A limited conception of what the equal sign means is one of the major stumbling blocks in learning algebra." Carpenter, Franke and Levi. It is necessary to build a cohesive understanding of equivalency, allowing students to manipulate equations and describe relationships.

Relational thinking is the ability to reason about mathematical relationships without always relying on computation. It involves recognizing patterns, using the properties of operations and understanding the structure of expressions to make judgements, justify solutions, and solve problems more flexibly and efficiently.

Structural Shifts

- The 2022 Standards use the term relational thinking to describe equivalency, emphasizing relationships between quantities rather than just computation.
- Many of the benchmarks in this anchor standard were in 2007's Algebra strand.
- This anchor standard highlights reasoning over computation. It focuses on students recognizing structure, mathematical properties, and using this to determine equivalency.
- The goal is to build toward Algebra with a deep understanding of the equal sign, enabling students to flexibly work with equations to solve problems, make conjectures, and justify their thinking.

Content Shifts

- Equality is defined as maintaining a balance on both sides of an equal sign. Students should not see an equal sign as something to do but as a symbol representing equal value on both sides.
- Students use equality with properties, place value, relationships, and/or structure to make conjectures and justify their reasoning.
- In middle and high school, students recognize and use the structure of number sentences, equations, and algebraic relationships to solve contextual problems and model situations.
- The standard includes a K-12 progression of comparing equivalent forms of expressions.

Strand 3: Patterns and Relationships

Mathematics is the science of pattern and order. Students will organize and describe the world using representations of numbers and operations. They will learn actions that transform numbers as well as ways of thinking that bring them back to where they started. Students will analyze and describe relationships among variables by focusing on how variables change and stay the same.

Anchor Standard 7: Patterns and Relationships

Represent and connect mathematical patterns and relationships using verbal descriptions, generalizations, tables, and graphs. Use representations to generate questions, make predictions, and solve mathematical problems.

“Effective teaching of mathematics engages students in making connections among mathematical representations to deepen understanding of mathematics concepts and procedures as tools for problem-solving.” NCTM's Principles to Action. Students’ ability to notice patterns and describe relationships throughout K-12 math ensures students’ mastery of algebra in the higher grades.

Structural Shifts

- The 2022 Patterns & Relationships strand combines content from the 2007 Number & Operation and Algebra strands into two new anchor standards. While Equivalence and Relational Thinking (Anchor Standard 6) includes content from both 2007 strands, most of what is traditionally thought of as algebra is included in Patterns and Relationships (Anchor Standard 7).
- A focus on patterns and relationships aligns with the definition of mathematics found in the Career, College and Community Readiness Statement. Math is the study of patterns and relationships. Students should have daily experiences with noticing patterns, describing patterns, and generalizing these relationships.

Content Shifts

- In grades K-5, students first experience algebra through the use of visual patterns.
- Students are given multiple opportunities throughout the year to notice and describe change.
- Students represent the patterns they see using expressions connected to contexts and visual patterns.
- Students engage in tasks that build algebraic habits of mind such as doing and undoing, abstracting from computation (MP7), and building rules to represent relationships and functions.
- Students connect multiple representations including contexts, language, equations, tables, and graphs.

Grade Level Shifts

This section summarizes high-level grade-level shifts between Minnesota's 2007 and 2022 K-12 Mathematics Standards, including notable content that moved grades or was omitted and new content added at each grade. This **non-exhaustive list** highlights the most significant shifts in math content by grade level.

Grade-level shifts fall into two categories: content adjustments and new content. Content adjustments include content from 2007 Standards that was removed or moved to a different grade in the 2022 Standards. New content includes content added in the 2022 Standards.

Kindergarten

Kindergarten benchmarks begin with the numeral zero (0). The following shifts fall into two categories. Content adjustments include content omitted or moved between grades, and new content includes content added in the 2022 Standards.

Content Adjustments

- Comparing and ordering whole numbers from 0 to 20 without objects

New Content

- Data Science is a new standard with 2 benchmarks.
- Counting by whole numbers extended from 20 to 31 to match the number of days in a month
- Counting using ten frames and other tools
- Fluently adding and subtracting within 5
- Skip counting by 10s from zero to 100
- Counting collections by grouping in 10s
- Composing and decomposing simple shapes

Grade 1

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Telling time to the hour and half-hour
- Explicit mention of finding 10 more or 10 less omitted and shifted to multiple new benchmarks in 2022 Standards (1.3.5.7, 1.3.5.2 and 1.3.7.2)

New Content

- Data Science and Chance & Uncertainty benchmarks
- Representing and solving contextual equal sharing situations where a whole is shared equally among 2 groups, and naming the fractional amount using the word "half"
- Describing positions of objects using left and right
- Solving addition within 100, including adding a two-digit number with a one-digit number and understanding when it is necessary to compose a new ten
- Defining 10, 20, 30, etc., as one, two, or three groups of 10s
- Representing numbers in expanded notation to 120
- Equal sharing situations with halves and fourths
- Describing changes in visual growing patterns

Grade 2

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Rounding whole numbers up/down to the nearest 10 and 100
- Telling time to the quarter-hour

New Content

- Data Science and Chance & Uncertainty benchmarks
- Visualizing data with dot plots and picture graphs
- Representing whole numbers as lengths from 0 on a number line
- Using $ and ¢ symbols when finding the value of coins
- Developing fluency in addition and subtraction within 20 using specific mental strategies (incrementing, compensation, fact families).
- Modeling sums and differences on a number line
- Using numeric expressions to describe visual growing patterns

Grade 3

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Determining elapsed time to the minute
- Using an analog thermometer (Fahrenheit/Celsius)
- Note: The use of time, clocks, and thermometers can be used as a context in the learning of other grade-level benchmarks.
- Rounding numbers to the nearest 10, 100, 1,000, and 10,000 to estimate sums and differences
- Identification of parallel and perpendicular lines (moved to grade 4)

New Content

- Data Science strand expanded (e.g. critically analyzing data visualizations to support a claim)
- Chance & Uncertainty benchmarks
- Mentally finding 100/1,000/10,000 more or less than a number
- Multiplication and division fluency within 144 (vs. basic facts in 2007)
- Financial literacy: addition and subtraction with estimated whole numbers to create short-term and long-term spending and saving goals
- Partitioning a whole into thirds and eighths
- Using relational thinking to find values in an open number sentence
- Representing non-unit fractions as sums of unit fractions
- Developing fluency specifically includes square products (e.g., 9×9)
- Measuring lengths to the nearest fourth when using standard units
- Using visual models to represent multiplication and division
- Finding previous terms in a visual pattern (growing/shrinking patterns)

Grade 4

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Geometric transformations (translations, rotations, and reflections) (moved to grade 7)
- Dividing by two-digit divisors (moved to grade 5)
- Using place-value to describe decimals to the thousandths (moved to grade 5)
- Reading and writing decimals up to thousandths
- Venn diagrams

New Content

- Data Science strand expanded (e.g., analyzing variability and addressing bias in data)
- Chance and Uncertainty benchmarks
- Using a number line to connect the probability values of 0 to impossible, 1/2 to equally likely, and 1 to certain
- Identifying parallel and perpendicular lines (moved from grade 3)
- Place value expanded to 1,000,000 for comparing and ordering numbers
- Fraction denominators include 2, 3, 4, 5, 6, 8, 10 and 12
- Recognizing that in a multi-digit whole number, a digit represents 10 times the value of the place to its right
- Using equations, rectangular arrays, and area models to represent and justify multiplication strategies
- Sorting and classifying quadrilaterals in a hierarchy (note: trapezoids have both inclusive and exclusive definition resulting in two hierarchies)
- Drawing nets of cubes
- Creating and describing different views of 3-dimensional figures
- Financial literacy: using the four operations to make financial decisions based on income, spending, saving, credit, and charitable giving
- Making change up to $20
- Measuring lengths to the nearest sixteenth of an inch and tenth of a centimeter
- Developing explicit rules from visual patterns
- Comparing and contrasting additive and multiplicative patterns

Grade 5

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Reading and writing decimals to the millionths
- Rounding decimals to the nearest 0.1, 0.01, or 0.001
- Using spreadsheets to find mean, median, and range

New Content

- Data Science strand expanded
- Comparing and contrasting qualitative and quantitative data
- Financial literacy: using the four operations to compare and contrast payment methods and explore models of debt and budgeting
- Chance and Uncertainty benchmarks (e.g., listing outcomes in frequency tables and placing predictions on a number line from 0 to 1)
- Dividing multi-digit whole numbers using two-digit divisors (moved from grade 4)
- Using place value to describe decimals to the thousandths (moved from grade 4)
- Drawing nets for prisms, pyramids, cylinders, and cones
- Justifying formulas for the area of parallelograms and triangles
- Representing contextual multiplication problems involving a fraction of a whole number using visual models, including number lines
- Using visual models to represent contextual measurement division problems and explaining the result
- Using ratio tables and interpreting multiplication as scaling

Grade 6

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Justifying fraction division through inverse operations (multiplication)
- Estimating the perimeter and area of irregular figures on a grid
- Solving problems involving angles formed by intersecting lines
- Explicit use of angle terms (e.g., supplementary, complementary, exterior, vertical) (moved to high school)

New Content

- Data Science benchmarks
- Measures of variability: interquartile range and mean-absolute deviation (MAD)
- Creating stem-and-leaf plots and box plots to describe data distribution
- Determining theoretical probability
- Finding the volume of nonrectangular prisms
- Drawing polygons on a coordinate plane and finding the lengths of sides
- Using the definition of the absolute value as a distance from zero (moved from grade 7)
- Identifying positive and negative numbers and plotting them on a coordinate grid
- Multiplying and dividing fractions and mixed numbers using visual models to represent the situation
- Solving one-step equations involving positive rational numbers
- Modeling ratios and percents with tables of equivalent ratios, tape diagrams, and double number lines

Grade 7

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Introduction and initial work with negative rational numbers (moved to grade 6)
- Absolute value as a distance from zero (moved to grade 6)
- Compound interest calculations (moved to grade 8). Note: simple interest calculations remain in grade 7.
- Distinction between proportional relationships and inversely proportional relationships
- Calculator truncation and rounding numbers

New Content

- Creating inferences about a population based on random samples and evaluating validity of generalizations
- Using measures of center and variability from random samples to draw informal comparative inferences about two populations
- Representing sample spaces for compound events using organized lists, tables, and tree diagrams (moved from grade 6)
- Designing and using simulations within computational tools (spreadsheets or calculators) to generate frequencies for compound events
- Transformations: translations and reflections (moved from grade 4)
- Financial literacy: creating a budget using positive and negative rational numbers and calculating percentage allocations
- Using models such as tables of equivalent ratios or double number lines

Grade 8

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Scientific notation using the correct number of significant digits

New Content

- Constructing and interpreting scatter plots to identify patterns such as clustering, outliers, and nonlinear association
- Communicating competing explanations based on correlation versus causation in data analysis
- Assessing model fit by judging the closeness of data points to the line
- Using similar triangles to explain why the slope (m) is constant on a non-vertical line and deriving the linear equation y=mx+b
- Recognizing f(x) =mx+b is a translation of y = mx
- Financial literacy: comparing and contrasting employment payment methods (per hour, salary, piecework) and justifying decisions with representations
- Compound interest calculations (moved from grade 7)

High School

High school benchmarks are grade-banded for grades 9-11, and all benchmark codes begin with the numeral nine (9). Districts should organize high school math benchmarks into three year-long courses, with approximately one-third of the benchmarks aligned to each course.

The following shifts fall into two categories. Content adjustments include content that was either omitted or moved to another grade level. New content includes content added in the 2022 Standards.

Content Adjustments

- Expressing proofs in specific forms (two-column proofs, paragraph proofs, flow charts)
- Dividing a polynomial by another polynomial
- Solving linear programming problems graphically
- Geometric transformations (moved to grade 7 and 8 as transformations of slope triangles)
- Law of large numbers
- Concept of an asymptote

New Content

- Data & Probability benchmarks increased to nearly a third of all benchmarks
- Identifying, creating, and comparing statistical models (linear/exponential) and assessing model fit using residuals
- Constructing and interpreting two-way frequency tables to calculate conditional probabilities and determine if events are independent
- Calculating and making decisions based on expected values using theoretical or experimental probabilities
- Comparing the recursion for Pascal's triangle, coin flips, and subsets
- Constructing logical arguments as proofs
- Using geometric transformations (rigid and non-rigid) to prove congruence
- Financial literacy topics integrated throughout (e.g., loan payoff time)
- Function composition and finding domain/range of advanced functions (piecewise and step functions)
- Using dimensional analysis to solve multi-step situations
- Using matrices to represent and manipulate data